Now $\underline{n=4}$: $A_4 = -\frac{1}{4}\sum_{k_1k_2k_3k_4} \mathrm{tr}\left\{ G_{k_1}\ \hat{\sigma}_3 \begin{pmatrix} 0 & B_{k_1k_2} \\ \bar{B}_{k_1k_2} & 0 \end{pmatrix} G_{k_2}\ \hat{\sigma}_3 \begin{pmatrix} 0 & B_{k_2k_3} \\ \bar{B}_{k_2k_3} & 0 \end{pmatrix} \right.$

$$\left. \cdot\, G_{k_3}\ \hat{\sigma}_3 \begin{pmatrix} 0 & B_{k_3k_4} \\ \bar{B}_{k_3k_4} & 0 \end{pmatrix} G_{k_4}\ \hat{\sigma}_3 \begin{pmatrix} 0 & B_{k_4k_1} \\ \bar{B}_{k_4k_1} & 0 \end{pmatrix} \right\} \left(\frac{1}{\sqrt{\beta V}}\right)^4 =$$

$$= -\frac{1}{4(\beta V)^2}\sum_{(k_i)} G_{k_1}G_{k_2}G_{k_3}G_{k_4}\,\mathrm{tr}\left[\begin{pmatrix} 0 & B_{k_1k_2} \\ -\bar{B}_{k_1k_2} & 0 \end{pmatrix}\begin{pmatrix} 0 & B_{k_2k_3} \\ -\bar{B}_{k_2k_3} & 0 \end{pmatrix}\begin{pmatrix} 0 & B_{k_3k_4} \\ -\bar{B}_{k_3k_4} & 0 \end{pmatrix}\begin{pmatrix} 0 & B_{k_4k_1} \\ -\bar{B}_{k_4k_1} & 0 \end{pmatrix}\right]$$

$$= -\frac{1}{4(\beta V)^2}\sum G_1G_2G_3G_4\,\mathrm{tr}\left[\begin{pmatrix} -B_{12}\bar{B}_{23} & 0 \\ 0 & -\bar{B}_{12}B_{23} \end{pmatrix}\begin{pmatrix} -B_{34}\bar{B}_{41} & 0 \\ 0 & -\bar{B}_{34}B_{41} \end{pmatrix}\right] =$$

$$= -\frac{1}{4(\beta V)^2}\sum G_1G_2G_3G_4\,\mathrm{tr}\begin{pmatrix} B_{12}\bar{B}_{23}B_{34}\bar{B}_{41} & 0 \\ 0 & \bar{B}_{12}B_{23}\bar{B}_{34}B_{41} \end{pmatrix}$$

$$= -\frac{1}{4(\beta V)^2}\sum_{1234} G_1G_2G_3G_4\left(B_{12}\bar{B}_{23}B_{34}\bar{B}_{41} + h.c.\right)$$

$$B_{12} = \sum_{q_1} \delta_{1+2,q_1}\,\phi_{q_1}$$

$$= -\frac{1}{4(\beta V)^2}\sum_{1234}\ \sum_{q_1q_2q_3q_4} \delta_{1+2,q_1}\,\delta_{2+3,q_2}\,\delta_{3+4,q_3}\,\delta_{4+1,q_4}$$

$$\cdot\, G_1G_2G_3G_4\left(\phi_{q_1}\bar{\phi}_{q_2}\phi_{q_3}\bar{\phi}_{q_4} + h.c.\right)$$

$k_2 = q_1 - k_1$; $k_3 = q_2 - k_2 = q_2 - q_1 + k_1$;

$k_4 = q_3 - k_3 = q_3 - q_2 + q_1 - k_1$;

and $k_4 + k_1 = q_4 = q_3 - q_2 + q_1$

$$A_4 = -\frac{1}{4}\sum_{k_1 q_1 q_2 q_3} G_{k_1} G_{q_1-k_1} G_{q_2-q_1+k_1} G_{q_3-q_2+q_1-k_1} \cdot$$
$$\cdot \left(\phi_{q_1} \bar{\phi}_{q_2} \phi_{q_3} \bar{\phi}_{q_4} + h.c.\right)$$

The condensate takes place at $\underline{\underline{q_i = 0}} \Rightarrow \phi_{q_i} = \Delta\, \delta_{q_i, 0}$

$$A_4 = -\frac{1}{4\beta V^2}(2) \sum_{k_1} G_{k_1} G_{-k_1} G_{k_1} G_{-k_1} \left(\Delta \bar{\Delta}\right)^2 =$$

$$= -\frac{1}{2\beta V^2} |\Delta|^4 \sum_k G_k^2 G_{-k}^2 = -\frac{1}{2\beta V} |\Delta|^4 \frac{1}{\beta V} \sum_k \left(G_k G_{-k}\right)^2;$$

$$\Rightarrow \frac{1}{\beta V} \sum_k G_k^2 G_{-k}^2 = \frac{1}{\beta} \sum_\omega \frac{1}{V} \sum_{\vec{k}} \frac{1}{(i\omega - \xi_k)^2} \frac{1}{(-i\omega - \xi_k)^2} =$$

$$= (T=0) = \int \frac{d\omega}{2\pi} \int \frac{d^d \vec{k}}{(2\pi)^d} \frac{1}{(\omega^2 + \xi_k^2)^2} = \frac{1}{4} \int \frac{d^d \vec{k}}{(2\pi)^d} \frac{1}{\xi_k^3}.$$

~~Thus we obtain $Z = e^{-S_{eff}} = e^{-\frac{1}{g}|\Delta|^2 A}$~~
~~(selecting only $q=0$ modes here)~~

Focus on $q = (\omega_n, \vec{q}) = 0$ mode now:

$$Z = \int D\bar{\phi} D\phi \, \exp\left(-\left\{\frac{1}{g}|\Delta|^2 - \left(\frac{1}{\beta}\sum_\omega \int (d\vec{k}) \frac{1}{\omega_n^2 + \xi_k^2}\right)|\Delta|^2\right.\right.$$
$$\left.\left. + \frac{1}{2\beta V}|\Delta|^4 \cdot \frac{1}{\beta}\sum_\omega \int (d\vec{k}) \frac{1}{(\omega_n^2 + \xi_k^2)^2}\right\} \cdot \exp - S[\phi]$$

fluctuational part of action, includes $\phi_{q \neq 0}$ modes.

Now we re-define condensate

$$\Delta = \sqrt{V}\,\Delta_0 \quad , \text{ so that } |\Delta|^2 = V|\Delta_0|^2$$
$$\text{and } \frac{1}{V}|\Delta|^4 = V|\Delta_0|^4$$

both terms are extensive (proportional to volume V) now.

$$Z = Z_0 \, Z_\phi$$

$$Z_0 = \exp(-V)\left\{ |\Delta_0|^2 \left(\frac{1}{g} - \int (d\omega_n d\vec{k}) \frac{1}{\omega_n^2 + \xi_k^2} \right) + \right.$$
$$\left. + \frac{1}{2\beta} \int (d\omega_n d\vec{k}) \frac{1}{(\omega_n^2 + \xi_k^2)^2} \cdot |\Delta_0|^4 \right\}$$

Here we denote $\int (d\omega_n d\vec{k}) \ldots \equiv \frac{1}{\beta} \sum_{\omega_n} \int \frac{d^d\vec{k}}{(2\pi)^d} \ldots$

Quadratic term: $|\Delta_0|^2 \left(\frac{1}{g} - L(0,0)\right)$

$$L(0,0) = \frac{1}{\beta} \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^3} \frac{1}{\omega_n^2 + \xi_k^2} = N(0) \ln\left(\frac{2e^{\gamma_E}}{\pi} \frac{\omega_D}{T}\right)$$

$\Rightarrow$ re-write $\frac{1}{g} - L(0,0) = N(0) \ln\left(\frac{T}{T_c}\right)$, where

$$N(0) \ln \frac{1}{T_c} = N(0)\left(\frac{1}{g N(0)} + \ln \frac{\pi}{2e^{\gamma}\omega_D}\right)$$

$$\Rightarrow \ln \frac{\pi T_c}{2e^{\gamma}\omega_D} = -\frac{1}{g N(0)}$$

$$T_c = \frac{2e^{\gamma_E}}{\pi} \omega_D e^{-\frac{1}{gN(0)}}$$ {superconducting critical temperature T_c

Next, $\ln \frac{T}{T_c} = \ln \frac{T_c + T - T_c}{T_c} = \ln\left(1 + \frac{T-T_c}{T_c}\right) \approx \frac{T-T_c}{T_c}$ for $T - T_c \ll T_c$

Quartic term $|\Delta_0|^4 \frac{1}{2\beta} \sum_{\omega} \int \frac{d\vec{k}}{(2\pi)^3} \frac{1}{(\omega_n^2 + \xi_k^2)^2} =$

$$\cong |\Delta_0|^4 \frac{1}{\beta} \sum_{\omega_n > 0} N(0) \int_{-\infty}^{\infty} d\xi \frac{1}{(\omega_n^2 + \xi^2)^2} =$$

$$= |\Delta_0|^4 \frac{7\zeta(3)}{16\pi^2} \beta^2 N(0)$$

because $\int_{-\infty}^{\infty} d\xi \frac{1}{(\omega_n^2 + \xi^2)^2} = \frac{\pi}{2\omega_n^3}$

$$\frac{1}{\beta} \sum_{\omega_n > 0} \frac{1}{\omega_n^3} = \frac{\beta^2}{\pi^3} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} = \frac{7}{8}\zeta(3) \frac{\beta^2}{\pi^3}$$

$$Z_0 = e^{-\beta F_0} \Rightarrow \frac{\beta F_0}{V} = N(0)\,\frac{T-T_c}{T_c}\,|\Delta_0|^2 +$$

$$+ \frac{7\zeta(3)}{8\pi^2}\,\frac{N(0)}{T_c^2}\,|\Delta_0|^4$$

is Landau-Ginzburg free energy for uniform superconducting condensate.

Observe that for $T > T_c$ the minimum of F_0 is at $\Delta_0 = 0$ (disordered phase).

For $T < T_c$ non-trivial minimum appears:

$$\frac{\partial F_0}{\partial \Delta_0} = 0 \Rightarrow \frac{T-T_c}{T_c}\cdot 2\Delta_0 + \frac{7\zeta(3)}{8\pi^2}\,\frac{4}{T_c^2}\,\Delta_0^3 = 0$$

$$\Delta_0^2 = \frac{2\pi^2}{7\zeta(3)}\,T_c\,(T_c - T)$$

$$\Delta_0 = \sqrt{\frac{2\pi^2}{7\zeta(3)}}\,T_c\,\sqrt{\frac{T_c - T}{T_c}} \quad \text{for } T < T_c$$

$$\Delta_0 = 0 \quad \text{for } T > T_c.$$

We obtained superconducting transition, its T_c and the theory for the order parameter $\phi_q(\omega)$. The condensate Δ_0 is just $\Delta_0 = \dfrac{\phi_{q=0}(\omega=0)}{\sqrt{V}}$.

More generally, $\phi_q(\omega) = \sqrt{V}\,\Delta_0\,\delta_{q,0}\,\delta_{\omega,0} + \tilde{\phi}_q(\omega)$ and superconducting fluctuations $\tilde{\phi}_q(\omega)$ are described by fluctuational part of the action.

$|\Delta|^2$ coefficient, including $\vec{q}$ and Ω_n dependence:

$$L(\vec{q},\Omega) = \frac{1}{\beta}\sum_{\omega_n}\int\frac{d\vec{k}}{(2\pi)^d}\,\frac{1}{(i\omega_n-\xi_{\vec{k}})(i\Omega-i\omega_n-\xi_{\vec{k}-\vec{q}})}$$

$$L(0,0) = \frac{1}{\beta}\sum_n\int\frac{d\vec{k}}{(2\pi)^d}\,\frac{1}{\omega_n^2+\xi_k^2} = \int\frac{d\vec{k}}{(2\pi)^d}\,\frac{th\left(\frac{\beta\xi_k}{2}\right)}{2\xi_k}$$

$\underline{d=3}$: $$\int\frac{d^3\vec{k}}{(2\pi)^3}\,\frac{th\left(\frac{\beta\xi_k}{2}\right)}{2\xi_k} = \frac{4\pi}{(2\pi)^3}\int\frac{dk}{d\xi_k}\,d\xi_k\,k^2\,\frac{th\left(\frac{\beta\xi_k}{2}\right)}{2\xi_k} \Rightarrow$$

$$\xi_k = \frac{k^2-k_F^2}{2m} \qquad \frac{d\xi_k}{dk} = \frac{k}{m}\approx\frac{k_F}{m}$$ because states near the Fermi surface are the most important ones.

$$\Rightarrow \frac{mk_F^2}{2\pi^2 k_F}\int_{-\omega_D}^{\omega_D}d\xi\,\frac{th(\beta\xi/2)}{2\xi} = \frac{mk_F}{2\pi^2}\int_0^{\omega_D}\frac{d\xi}{\xi}\,th\left(\frac{\beta\xi}{2}\right)$$

Integrate over the shell $(-\omega_D,\omega_D)$ in energy around the Fermi surface because this is where the attractive interaction between electrons is present.

To calculate the integral $I = \int_0^{\omega_D}\frac{d\xi}{\xi}\,th\left(\frac{\beta\xi}{2}\right) = \int_0^{\beta\omega_D/2}\frac{dx}{x}\,th\,x$

assume that $\beta\omega_D \gg 1$ (as we confirm soon).

Then $$I = \int_0^{\beta\omega_D/2}d(\ln x)\,th\,x \cong \ln x\,th\,x\Big|_0^{\beta\omega_D/2} - \int_0^\infty\frac{\ln x}{ch^2 x}\,dx =$$

$$= \ln\frac{\beta\omega_D}{2} - \left(-\gamma_E + \ln\frac{\pi}{4}\right) = \ln\left(\frac{4e^{\gamma_E}}{\pi}\,\frac{\beta\omega_D}{2}\right) =$$

$$= \ln\left(\frac{2e^{\gamma_E}}{\pi}\,\frac{\omega_D}{T}\right)$$ where $\gamma_E = -\int_0^\infty e^{-x}\ln x\,dx =$

$= 0.5772\ldots$

is Euler–Mascheroni constant.

We can obtain $\vec{q}, \Omega$ dependence by expanding

$L(\vec{q}, \Omega) - L(0,0)$.

Let us start with $L(\vec{q}=0, \Omega) - L(0,0) \equiv \Delta L_1 =$

$$= \frac{1}{\beta} \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{(i\omega - \xi_k)(i\Omega - i\omega - \xi_k)} - \frac{1}{\beta} \sum_n \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{(i\omega - \xi)(-i\omega - \xi)}$$

Since $L(\vec{q}=0, \Omega=0) - L(0,0) = 0$, we can extend integration over ξ_k to $\pm\infty$:

$$\Delta L_1 = \frac{1}{\beta} \sum_{\omega_n} \left(\frac{m k_F}{2\pi^2}\right) \left\{ \int_{-\infty}^{\infty} d\xi \frac{1}{(\xi - i\omega)(\xi + i\omega - i\Omega)} - \int_{-\infty}^{\infty} d\xi \frac{1}{\xi^2 + \omega_n^2} \right\}$$

$$\omega_n = \frac{\pi}{\beta}(2n+1), \quad \Omega = \frac{2\pi m}{\beta}$$

$$B_1 = \frac{1}{\beta} \sum_n \int_{-\infty}^{\infty} d\xi \frac{1}{(\xi - i\omega)(\xi + i\omega - i\Omega)} = \frac{1}{\beta} \sum_n \int d\xi \left(\frac{1}{\xi - i\omega} + \frac{1}{\xi + i\omega - i\Omega} \right) \cdot \frac{1}{2\xi - i\Omega}$$

Next observe that $\sum_n \frac{1}{\xi + i\omega - i\Omega} = \sum_n \frac{1}{\xi + \frac{\pi}{\beta}(2n+1-2m)} =$

$= (\text{shift } n \to n+m) = \sum_n \frac{1}{\xi + \frac{\pi}{\beta}(2n+1)} = \sum_n \frac{1}{\xi + i\omega_n}$

$$\Rightarrow B_1 = \frac{1}{\beta} \sum_n \int d\xi \left(\frac{1}{\xi - i\omega} + \frac{1}{\xi + i\omega} \right) \frac{1}{2\xi - i\Omega}$$

$$B_1 = \frac{1}{\beta} \sum_{\omega_n} \int d\xi \; \frac{2\xi}{2\xi - i\Omega} \; \frac{1}{\xi^2 + \omega_n^2}$$

$$\Delta L_1 = \underbrace{\left(\frac{m k_F}{2\pi^2}\right)} \; \frac{1}{\beta} \sum_{\omega_n} \int_{-\infty}^{\infty} d\xi \; \frac{1}{\xi^2 + \omega_n^2} \left(\frac{2\xi}{2\xi - i\Omega} - 1 \right) =$$

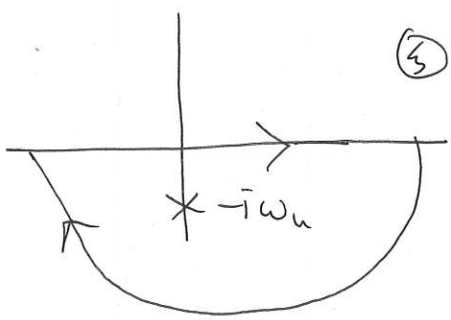
$$= \underbrace{N(0)}_{\text{density of states}} \; \frac{1}{\beta} \sum_{\omega_n} \int_{-\infty}^{\infty} d\xi \; \frac{i\Omega}{2\xi - i\Omega} \; \frac{1}{\xi^2 + \omega_n^2}$$

Need to consider $\Omega > 0$, $\Omega < 0$ separately.

<u>let $\Omega > 0$</u> : observe that

$$\frac{1}{\beta} \sum_{\omega_n} \frac{1}{\xi^2 + \omega_n^2} = \frac{2}{\beta} \sum_{\omega_n > 0} \frac{1}{\xi^2 + \omega_n^2}$$

$$\Delta L_1 = 2N(0) \; \frac{1}{\beta} \sum_{\omega_n > 0} \int_{-\infty}^{\infty} d\xi \; \frac{i\Omega}{(2\xi - i\Omega)(\xi - i\omega_n)(\xi + i\omega_n)}$$



$$\Delta L_1 = 2N(0) \; \frac{1}{\beta} \sum_{\omega_n > 0} \frac{i\Omega \, (-2\pi i)}{(-2i\omega_n - i\Omega)(-2i\omega_n)} =$$

$$= -4\pi N(0) \; \frac{1}{\beta} \sum_{\omega_n > 0} \frac{\Omega}{(2\omega_n + \Omega)(2\omega_n)} =$$

$$= -\pi N(0) \, (\Omega) \; \frac{1}{\beta} \sum_{\omega_n > 0} \frac{1}{\omega_n \left(\omega_n + \frac{\Omega}{2}\right)}$$

It is now safe to expand in $\frac{\Omega}{2\omega_n}$.

The leading term

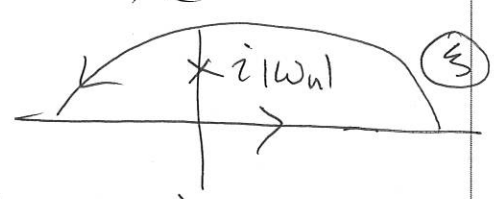
$$\Delta L_1 = -\pi N(0)\,(\Omega)\,\underbrace{\frac{1}{\beta}\sum_{\omega_n>0}\frac{1}{\omega_n^2}}_{T\sum_{n=0}^{\infty}\frac{1}{(\pi T(2n+1))^2}=}$$

$$= \frac{1}{\pi^2 T}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}$$

<u>Now let $\Omega = -|\Omega| < 0$:</u>

$$\frac{1}{\beta}\sum_{\omega_n}\frac{1}{\xi^2+\omega_n^2} = \frac{1}{\beta}\sum_{\omega_n}\frac{1}{\xi^2+|\omega_n|^2} = \frac{2}{\beta}\sum_{\omega_n>0}\frac{1}{\xi^2+|\omega_n|^2}$$

$$\Delta L_1 = 2N(0)\,\frac{1}{\beta}\sum_{\omega_n>0}\int d\xi\,\frac{-i|\Omega|}{(2\xi+i|\Omega|)(\xi+i|\omega_n|)(\xi-i|\omega_n|)} =$$



$$= 2N(0)\,\frac{1}{\beta}\sum_{\omega_n>0}(-i|\Omega|)\,\frac{2\pi i}{(2i|\omega_n|+i|\Omega|)\,2i|\omega_n|} =$$

$$= 2N(0)\,\frac{1}{\beta}\sum_{\omega_n>0}\frac{2\pi|\Omega|}{-4|\omega_n|\left(|\omega_n|+\frac{|\Omega|}{2}\right)} =$$

$$= -\pi N(0)\,|\Omega|\,\frac{1}{\beta}\sum_{\omega_n>0}\frac{1}{\omega_n^2\left(1+\frac{|\Omega|}{2|\omega_n|}\right)} \approx -\pi N(0)\,(|\Omega|)\cdot$$

$$\cdot\,\frac{1}{\beta}\sum_{\omega_n>0}\frac{1}{\omega_n^2}.$$

Thus $\Delta L_1 = -\pi N(0) \left(\frac{1}{\beta} \sum_{\omega_n > 0} \frac{1}{\omega_n^2} \right) |\Omega|$

This is valid for arbitrary sign of Ω.

$$\Delta L_1 = -\frac{\cancel{\pi} N(0) \beta}{\pi} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} |\Omega| = -\frac{\pi N(0)}{8 T_c} |\Omega|$$

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{3}{4} \zeta(2) = \frac{\pi^2}{8}$$, where $\zeta(k)$ = Riemann's Zeta function.

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} = \frac{7}{8} \zeta(3).$$

Now $\Delta L_2 = L(\vec{q},0) - L(0,0) = \frac{1}{\beta}\sum_{\omega_n}\int\frac{d\vec{k}}{(2\pi)^d}$ ~~[illegible]~~

$$\left(\frac{1}{(i\omega-\xi_k)(-i\omega-\xi_{k-q})} - \frac{1}{\omega_n^2+\xi_k^2}\right) =$$

$$= \frac{1}{\beta}\sum_{\omega_n}\int(dk)\,\frac{1}{(i\omega-\xi_k)}\left(\frac{1}{-i\omega-\xi_{k-q}} - \frac{1}{-i\omega-\xi_k}\right)$$

$$\frac{1}{-i\omega-\xi_{k-q}} = \frac{-1}{i\omega+\xi_{k-q}} = \frac{-1}{i\omega+\xi_k+\delta\xi_k} = \frac{-1}{(i\omega+\xi_k)}\left(1-\frac{\delta\xi_k}{i\omega+\xi_k}+\frac{\delta\xi_k^2}{(i\omega+\xi_k)^2}+\dots\right)$$

$$\xi_{k-q} = \frac{(\vec{k}-\vec{q})^2-k_F^2}{2m} = \underbrace{\frac{k^2-k_F^2}{2m}}_{\xi_k} \underbrace{-\frac{\vec{k}\cdot\vec{q}}{m}+\frac{q^2}{2m}}_{\delta\xi_k}$$

$$\Delta L_2 = \frac{1}{\beta}\sum_{\omega}\int(d\vec{k})\,\frac{1}{i\omega-\xi_k}\left(\frac{\delta\xi_k}{(i\omega+\xi_k)^2} - \frac{\delta\xi_k^2}{(i\omega+\xi_k)^3}\right)$$

Work to q^2-order $\Rightarrow$ $\delta\xi_k^2 = \left(\frac{\vec{k}\cdot\vec{q}}{m}\right)^2 = \frac{k^2q^2}{m^2}\cos^2\theta$

$$\delta\xi_k = \frac{q^2}{2m} - \frac{kq}{m}\cos\theta$$

Choose z-axis along $\vec{q}$, then

$$\int(d\vec{k}) = \int_0^\infty\frac{dk\,k^2}{(2\pi)^3}\int_0^{2\pi}d\varphi\int_0^\pi d\theta\sin\theta \cong \frac{k_F^2}{2\cdot 2\pi^2}\int dk\int_0^\pi d\theta\sin\theta$$

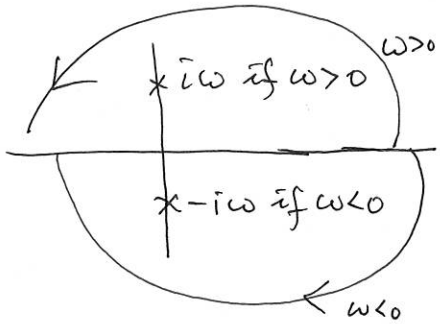
Hence $\int_0^\pi d\theta\sin\theta\cos\theta = 0$ and only $\left(\frac{q^2}{2m}\right)$ contributes in $\delta\xi_k$.

In $\delta\xi_k^2$: $\int_0^\pi d\theta\sin\theta\cos^2\theta = \frac{2}{3}$.

$$\Delta L_2 = \frac{1}{\beta}\sum_{\omega} \int_0^\infty dk \frac{k_F^2}{2\pi^2}\left(\frac{\frac{q^2}{m}}{(i\omega+\xi_k)^2} - \frac{\frac{1}{2}\cdot\frac{2}{3}\frac{q^2}{m^2}k^2}{(i\omega+\xi_k)^3}\right)\frac{1}{i\omega-\xi_k}$$

$$\approx \frac{1}{\beta}\sum_{\omega}\frac{(mk_F)}{2\pi^2}\int_{-\infty}^{\infty} d\xi_k \left\{\frac{q^2/m}{(i\omega-\xi_k)(i\omega+\xi_k)^2} - \frac{q^2k_F^2}{3m^2}\cdot\frac{1}{(i\omega+\xi_k)^3}\cdot\frac{1}{(i\omega-\xi_k)}\right\}$$

$$\int_{-\infty}^{\infty} d\xi \frac{1}{(i\omega-\xi)(i\omega+\xi)^p} = -\int_{-\infty}^{\infty} d\xi \frac{1}{(\xi+i\omega)^p(\xi-i\omega)}$$



$$= (-)\left(\frac{2\pi i\,\theta(\omega)}{(i\omega)^p} + \frac{-2\pi i\,\theta(-\omega)}{(i\omega)^p}\right)$$

$$\cancel{\sum_{\omega}\frac{1}{(i\omega-\xi)(i\omega+\xi)^p} = \sum_{\omega>0} + \sum_{\omega<0}}$$

$$\cancel{\sum_{\omega<0} = \sum_{\omega'=-\omega}\frac{1}{(-i\omega'-\xi)(\xi-i\omega')^p}}$$

But $$\sum_{\omega_n}\frac{\theta(\omega_n)}{(i\omega_n)^p} = \sum_{\omega_n>0}\frac{1}{(i\omega_n)^p}$$

$$\sum_{\omega_n}\frac{\theta(-\omega_n)}{(i\omega_n)^p} = \sum_{\omega_n<0}\frac{1}{(i\omega_n)^p} = \begin{pmatrix}\omega_n=-\omega_n' \\ \omega_n'>0\end{pmatrix} =$$

$$= \sum_{\omega_n'>0}\frac{1}{(-i\omega_n')^p} = (-1)^p\sum_{\omega_n>0}\frac{1}{(i\omega_n)^p}$$

Thus for $\underline{p=2}$ we have

$$\frac{1}{\beta}\sum_{\omega_n} \frac{2\pi i}{(i\omega_n)^2}\left(\theta(\omega_n) - \theta(-\omega_n)\right) = \frac{1}{\beta}\sum_{\omega_n>0} \frac{2\pi i}{(i\omega_n)^2}\left(1-(-1)^2\right) = 0$$

While for $\underline{p=3}$ we get

$$\frac{1}{\beta}\sum_{\omega_n>0} \frac{2\pi i}{(i\omega_n)^3}\left(1-(-1)^3\right) = \frac{1}{\beta}\sum_{\omega_n>0} \frac{-4\pi}{\omega_n^3}$$

$$\Rightarrow \Delta L_2 = \frac{(mk_F)}{2\pi^2}\frac{q^2 k_F^2}{3m^2}(-4\pi)\frac{1}{\beta}\sum_{\omega_n>0}\frac{1}{\left(\frac{\pi}{\beta}(2n+1)\right)^3} =$$

$$= -\frac{1}{3}q^2\frac{k_F^3\beta^2}{m\cdot 2\pi^2}\frac{4\pi}{\pi^3}\underbrace{\sum_{n=0}^{\infty}\frac{1}{(2n+1)^3}}_{\frac{7}{8}\zeta(3)} =$$

$$= -\frac{7}{6\pi^2}\zeta(3)\frac{k_F^3\beta^2}{2\pi^2 m}q^2 = -\frac{7}{6\pi^2}\zeta(3)\,N(0)\,\beta^2 v_F^2 q^2$$

In terms of density of states $N(0) = \frac{mk_F}{2\pi^2}$:

$$\frac{k_F^3}{2\pi^2 m} = \frac{mk_F}{2\pi^2}\frac{k_F^2}{m^2} = N(0)\,v_F^2 .$$

$$\Delta L_2 = -N(0)\,\frac{7\zeta(3)}{6\pi^2}\beta^2 v_F^2 q^2 .$$

More general form of $S[\phi]$ action of superconducting fluctuations:

$$Z = \int D\bar{\phi}\, D\phi \, \exp\Big\{ - \sum_{q=(\vec{q},\Omega)} \bar{\phi}_q \phi_q \Big(\frac{1}{g} - L(\vec{q},\Omega) \Big)$$

$$- \frac{u}{2} \sum_{q_1 q_2 q_3} \big(\bar{\phi}_{q_1} \phi_{q_2} \bar{\phi}_{q_3} \phi_{-q_1-q_2-q_3} + h.c. \big) \Big\}$$

where $L(\vec{q},\Omega) = L(0,0) - \frac{\pi N(0)}{8T_c} |\Omega| - \frac{7\zeta(3) N(0)}{6\pi^2 T_c^2} v_F^2 q^2$

$$u = \frac{7\zeta(3)}{16\pi^2 T_c^2} N(0)$$

Non-condensate part of $\phi_{q\neq 0}$ is dynamical interacting quantum field $\rightarrow$ 2-component ϕ^4-theory.